

- Open

Git bash shell and create a new folder “**GitDemo**”
and initialize as git repository

```
surya.p@PTPLL487 MINGW64 ~/Desktop/GitDemo  
$ git init  
Initialized empty Git repository in C:/Users/surya.p/Desktop/GitDemo/.git/
```

- Git

bash initializes the “**GitDemo**”
repository.

```
surya.p@PTPLL487 MINGW64 ~/Desktop/GitDemo (master)  
$ git remote add origin https://github.com/SURYA1804/GitDemo.git
```

- Create

a file “**welcome.txt**”
and add content to the file (use echo or cat command)

```
surya.p@PTPLL487 MINGW64 ~/Desktop/GitDemo (main)  
$ echo "Welcome to world of mine" > welcome.txt
```

- To

verify if the file “welcome.txt” is created, execute ls-al

```
surya.p@PTPLL487 MINGW64 ~/Desktop/GitDemo (main)
$ ls
sample/  welcome.txt
```

- To

verify the content, execute the command cat filename

```
surya.p@PTPLL487 MINGW64 ~/Desktop/GitDemo (main)
$ cat welcome.txt
Welcome to world of mine
```

- Execute

git status

```
surya.p@PTPLL487 MINGW64 ~/Desktop/GitDemo (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore

nothing added to commit but untracked files present (use "git add" to track)
```

- Signup

with GitHub and create a remote repository “**GitDemo**”

- Add

remote origin

```
surya.p@PTPLL487 MINGW64 ~/Desktop/GitDemo (master)
$ git remote add origin https://github.com/SURYA1804/GitDemo.git
```

- Create

files and push the local to remote repository.

```
surya.p@PTPLL487 MINGW64 ~/Desktop/GitDemo (main)
$ git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 236 bytes | 78.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/SURYA1804/GitDemo.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

- Make

changes in the remote and pull the changes to the local.

```
surya.p@PTPLL487 MINGW64 ~/Desktop/GitDemo (main)
$ git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 937 bytes | 44.00 KiB/s, done.
From https://github.com/SURYA1804/GitDemo
   f2d6bf0..9d865a8  main       -> origin/main
Updating f2d6bf0..9d865a8
Fast-forward
 welcome.txt | 1 +
 1 file changed, 1 insertion(+)
```

- Create

a .gitignore file to ignore contents of temp folder

```
surya.p@PTPLL487 MINGW64 ~/Desktop/GitDemo (main)
$ echo "sample/" > .gitignore
```

Docker

- What is Docker-compose.yaml file? Give sample docker-compose file.

A `docker-compose.yaml` (or `docker-compose.yml`) file is a configuration file used by Docker Compose—a tool for defining and running multi-container Docker applications. This YAML file specifies how different containers (called “services”) interact, what images they use, what ports to expose, environment variables, volumes, networks, and other configuration settings—all in a single place.

db:

image: postgres:latest

database

environment:

POSTGRES_PASSWORD: example variable

networks:

- my_network

volumes:

- db_data:/var/lib/postgresql/data

- What is Docker hub?

Docker Hub is the world's largest cloud-based container registry platform. It is designed to help developers store, manage, discover, and share Docker container images. Docker Hub serves as a central repository for container images, making it easy to collaborate on application development and deployment in modern CI/CD workflows.